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PROFESSIONAL SUMMARY

Generative AI-focused Data Scientist specializing in building production-grade LLM applications, Retrieval-Augmented Generation (RAG) systems, and intelligent AI assistants. Strong expertise in designing scalable, secure, and context-aware NLP solutions using LangChain, Hugging Face, and FastAPI. Proven ability to design end-to-end AI systems with efficient retrieval, reduced hallucination, and real-world business impact.

EXPERIENCE

Data Scientist | J.P. Morgan | Aug 2024 – Present | Hyderabad

- Architected and deployed an enterprise-grade HRMS chatbot powered by LLMs handling employee queries across payroll, leave, and policies.
- Designed a scalable RAG pipeline using embeddings and FAISS (vector database) to integrate structured HR databases and unstructured knowledge bases.
- Applied prompt engineering and LLM evaluation techniques to improve response relevance and reduce hallucinations.
- Built secure conversational workflows with role-based access control for sensitive HR data.
- Developed high-performance FastAPI services enabling real-time inference and system integration.
- Optimized retrieval accuracy and reduced latency through embedding tuning, indexing strategies, and caching.

Data Scientist | Inndata Analytics | Oct 2022 – Jul 2024 | Visakhapatnam

- Built an LLM-powered SQL ChatMate system converting natural language queries into optimized SQL.
- Designed NLP pipelines for intent understanding, schema mapping, and structured query generation.
- Integrated LLM outputs with relational databases for automated execution and insights generation.
- Applied prompt engineering, output validation, and error handling to improve accuracy and reliability.
- Developed scalable FastAPI-based APIs for deployment and real-time query processing.
- Built a face recognition-based attendance system (SPOT-SIGN) using TensorFlow and OpenCV for real-time employee tracking.
- Implemented face embeddings and similarity thresholding for accurate identity matching and authorization.
- Integrated the system with backend databases for automated attendance recording.

Data Science Intern | SocialTek AI ML Pvt Ltd | Feb 2022 – Sep 2022 | Hyderabad

- Developed a property price prediction model using Random Forest with strong predictive performance.
 - Performed web scraping using Selenium and conducted EDA and preprocessing.
 - Improved data quality through missing value handling and feature engineering.
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GENERATIVE AI PROJECTS

HRMS Chatbot (LLM + RAG + FastAPI)

- Built a production-ready conversational AI system using LLMs, LangChain, and FAISS.
- Enabled context-aware multi-turn conversations with memory handling.
- Designed modular RAG architecture for efficient retrieval and response generation.
- Implemented security mechanisms for handling sensitive enterprise data.

SQL ChatMate (LLM + NLP)

- Developed a natural language to SQL system leveraging LLMs and prompt engineering.
 - Automated database querying with intelligent schema understanding.
 - Integrated APIs for real-time query execution and visualization.
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PROJECTS

Face Recognition Attendance System (SPOT-SIGN)

- Developed during tenure at Inndata Analytics as part of real-time ML application.
- Built using TensorFlow and OpenCV with face embeddings and similarity thresholding.
- Enabled automated attendance tracking with database integration.

Property Price Navigator

- Built a machine learning model using Random Forest for property price prediction.
 - Performed web scraping, EDA, and preprocessing for high-quality data pipelines.
 - Achieved strong predictive performance with high adjusted R^2 .
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SKILLS

- Programming: Python, SQL
- Generative AI: LLMs, RAG, LangChain, Prompt Engineering, AI Agents
- Machine Learning: Regression, Classification, Model Evaluation, Feature Engineering
- Deep Learning: Neural Networks, TensorFlow
- NLP: Conversational AI, Text-to-SQL, Multi-turn Systems
- Frameworks & Libraries: FastAPI, TensorFlow, FAISS, OpenCV
- Data & Visualization: EDA, Data Wrangling, Data Pipelines
- Tools: Hugging Face, Docker, Git
- Concepts: System Design, Scalable AI Systems, Model Optimization, Data Processing

EDUCATION

Bachelor of Science in Computer Science
Andhra University, Visakhapatnam | 2021

CERTIFICATIONS

- Advanced Data Science and AI
 - Big Data and Data Science Challenge
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KEY HIGHLIGHTS

- Strong focus on Generative AI and production-grade LLM applications
- Expertise in RAG architecture and prompt engineering
- Experience building scalable and secure AI systems
- End-to-end ownership from model development to deployment